Chapter 6 Exercises: Selected Answers and Solutions

1. xxx

2. Let X be the number of patients cured. Then

$$X \sim B\left(15, 0.5\right).$$

(a) Directly from tables,

$$P\left(X \leq 6\right) = 0.3036.$$

(b) We see from the tables

$$\begin{aligned} P\left(6 \leq X \leq 10\right) &= P\left(X \leq 10\right) - P\left(X \leq 5\right) \\ &= 0.9408 - 0.1509 \\ &= 0.7899. \end{aligned}$$

(c) The mean is

$$E\left(X\right) = \frac{15}{2} = 7.5$$

and the variance is

$$\mathrm{Var}\left(X\right) = 15 \times \frac{1}{2} \times \frac{1}{2} = \frac{15}{4}.$$

Hence the standard deviation is $\mathrm{SD}(X) = \sqrt{15}/2$.

3. If X is the number of incoming calls per minute, then

$$X \sim Po\left(5\right).$$

(a) We see

$$\begin{aligned} P\left(X = 2\right) &= \frac{5^2 e^{-5}}{2} \\ &= 0.08422. \end{aligned}$$

(b) Via tables or otherwise

$$\begin{aligned} P\left(X \geq 3\right) &= 1 - P\left(X \leq 2\right) \\ &= 1 - \left(P\left(X = 0\right) + P\left(X = 1\right) + P\left(X = 2\right)\right) \\ &= 1 - 0.1247 \\ &= 0.8753. \end{aligned}$$

(c)

$$\begin{aligned} P(X=0) &= \frac{5^0 e^{-5}}{0!} \\ &= e^{-5} \\ &= 0.00678. \end{aligned}$$

(d) If

$$Y = X_1 + X_2 + X_3 + X_4 + X_5$$

represents the number of calls in five minutes, where $X_i \sim Po(5)$, then

$$\begin{aligned} E(Y) &= 5+5+5+5+5 \\ &= 25. \end{aligned}$$

4. Let X be the number of times a blue face is rolled. Then $X \sim B\left(8, \frac{5}{6}\right)$.

(a) We have

$$\begin{aligned} E(X) &= 8 \times \frac{5}{6} = \frac{20}{3}, \\ \text{Var}(X) &= 8 \times \frac{5}{6} \times \frac{1}{6} = \frac{10}{9}. \end{aligned}$$

(b)

$$\begin{aligned} P(X=6) &= \binom{8}{6} \left(\frac{5}{6}\right)^6 \left(\frac{1}{6}\right)^2 \\ &= 28 \left(\frac{5}{6}\right)^6 \left(\frac{1}{6}\right)^2 \\ &= 0.260\,5. \end{aligned}$$

(c)

$$\begin{aligned} P(X>6) &= P(X=7) + P(X=8) \\ &= 7\left(\frac{5}{6}\right)^7 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^8 \\ &= 0.325\,6 + 0.232\,6 \\ &= 0.558\,2. \end{aligned}$$

5. xxx

6. xxx

7. xxx

8. The "30-year wind" refers to a wind velocity occurring on average only every 30 years.

   (a) If the return is $\tau = 30$, then on a given year let

   $$p = P\left(\text{exceeding wind}\right) = \frac{1}{30}.$$

   If X is the year of the first exceedence, then $X \sim Ge\left(p\right)$. So

   $$P\left(X = 7\right) = \left(\frac{29}{30}\right)^6 \frac{1}{30} = 0.027198.$$

   (b) If Y denotes the number of exceedences in the 7 years, then $Y \sim B\left(7, \frac{1}{30}\right)$, so

   $$\begin{aligned} P\left(Y \geq 1\right) &= 1 - P\left(Y = 0\right) \\ &= 1 - \left(\frac{29}{30}\right)^7 \\ &= 0.2112. \end{aligned}$$

   (c) Let V be the maximum annual wind velocity, we are told $V \sim Exp\left(\frac{1}{35}\right)$. So

   $$P\left(V > v_0\right) = \frac{1}{30}$$

   for some velocity v_0. Using the survivor function here

   $$P\left(V > v_0\right) = e^{-v_0/35} = \frac{1}{30}.$$

   Hence, on taking logs,

   $$\begin{aligned} v_0 &= 35 \log\left(30\right) \\ &= 119.04 \end{aligned}$$

   miles per hour.

(d) If in fact $v_0 = 101.18$ miles per hour, then

$$\begin{aligned} P(V > 101.18) &= e^{-101.18/35} \\ &= 0.0555. \end{aligned}$$

So then τ is approximately 18 years.

9. We have $X_1, X_2 \sim B(14, 0.3)$, we assume independently.

(i)

$$\begin{aligned} P(X_1 = 4) &= \binom{14}{4} 0.3^4 0.7^{10} \\ &= 0.229\,03. \end{aligned}$$

From tables

$$\begin{aligned} P(X_1 < 6) &= P(X_1 \leq 5) \\ &= 0.7805. \end{aligned}$$

Further,

$$\begin{aligned} P(2 < X_1 < 6) &= P(3 \leq X \leq 5) \\ &= P(X \leq 5) - P(X \leq 2) \\ &= 0.7805 - 0.1608 \\ &= 0.6197. \end{aligned}$$

(ii)

$$\begin{aligned} E(X_1) &= 14 \times 0.3 = 4.2, \\ \mathrm{Var}(X_1) &= 14 \times 0.3 \times 0.7 = 2.94 \end{aligned}$$

and so

$$SD(X_1) = 1.715.$$

$$\begin{aligned} E(X_1 + X_2) &= 4.2 + 4.2 = 8.4, \\ \mathrm{Var}(X_1 + X_2) &= 2.94 + 2.94 = 5.88 \end{aligned}$$

by independence, so $\mathrm{SD}(X_1 + X_2) = 2.425$. Now

$$\begin{aligned} E(X_1 X_2) &= E(X_1) E(X_2) \\ &= 4.2^2 \\ &= 17.64, \end{aligned}$$

again by independence. The final part here is harder: note

$$\begin{aligned} \mathrm{Var}(X_1 X_2) &= E\left((X_1 X_2 - 17.64)^2\right) \\ &= E\left(X_1^2 X_2^2\right) - 2 \times 17.64 E(X_1 X_2) + 17.46^2 \\ &= E\left(X_1^2\right) E\left(X_2^2\right) - 2 \times 17.64^2 + 17.64^2 \\ &= E\left(X_1^2\right)^2 - 17.64^2. \end{aligned}$$

The problem reduces to finding $E(X^2)$ when $X \sim B(n, p)$. Armed with the information that $E(X) = np$, we require

$$E\left(X^2\right) = \sum_{x=0}^{n} x^2 \binom{n}{x} p^x (1-p)^{n-x}$$

using our "Law of the Unconscious Statistician". Writing $x^2 = x(x-1) + x$, we can split the summation into two, so

$$\begin{aligned} E\left(X^2\right) &= \sum_{x=0}^{n} x(x-1) \binom{n}{x} p^x (1-p)^{n-x} + \sum_{x=0}^{n} x \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=2}^{n} \frac{n!}{(n-x)!\,(x-2)!} p^x (1-p)^{n-x} + E(X) \\ &= n(n-1) p^2 \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} (1-p)^{n-x} + np \\ &= n(n-1) p^2 + np, \end{aligned}$$

since the summation on the penultimate line is unity (put $k = x-2$ to see why). In this case

$$\begin{aligned} E\left(X_1^2\right) &= 14 \times 13 \times 0.3^2 + 4.2 \\ &= 20.58. \end{aligned}$$

Hence

$$\begin{aligned} \mathrm{Var}\left(X_1X_2\right) &= 20.58^2 - 17.64^2 \\ &= 112.37. \end{aligned}$$

So $\mathrm{SD}(X_1X_2) = 10.60$.

(iii) Now $Y := X_1 + X_2 \sim B\left(28, 0.3\right)$, so

$$\begin{aligned} P\left(Y = 8\right) &= \binom{28}{8} 0.3^8 0.7^{20} \\ &= 0.162\,7. \end{aligned}$$

Further, via tables we find

$$\begin{aligned} P\left(Y < 12\right) &= P\left(Y \leq 11\right) \\ &= 0.89717. \end{aligned}$$

Finally, again using tables

$$\begin{aligned} P\left(4 < Y < 12\right) &= P\left(Y \leq 11\right) - P\left(Y \leq 4\right) \\ &= 0.89717 - 0.047427 \\ &= 0.84974. \end{aligned}$$

10. .

(a) 40

(b) 4 or 5

(c) 0.133

(d) 0.1245

(e) 1

11. We note that the probability of a red ball on each selection is 4/10.

(a) Here $X \sim B\left(100, \frac{2}{5}\right)$, so

$$P\left(X = k\right) = \binom{100}{k} \left(\frac{2}{5}\right)^k \left(\frac{3}{5}\right)^{100-k}$$

for $k = 0, 1, 2, \ldots, 100$.

(b) Hence

$$\begin{aligned} E(X) &= np = 100 \times \frac{2}{5} = 40, \\ \text{Var}(X) &= npq = 40 \times \frac{3}{5} = 24. \end{aligned}$$

(c) Via tables or computer software,

$$\begin{aligned} P(X \leq 38) &= \sum_{k=0}^{38} \binom{100}{k} \left(\frac{2}{5}\right)^k \left(\frac{3}{5}\right)^{100-k} \\ &= 0.382\,2. \end{aligned}$$

12. .

(a) 4, 4

(b) 0.135

(c) 5, 1.25

13. .

(a) xxx

(b) xxx

14. Let X be the number of correct answers. Then $X \sim B\left(10, \frac{1}{5}\right)$.

(a) From tables or directly,

$$\begin{aligned} P(X \leq 3) &= \sum_{x=0}^{3} \binom{10}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{10-x} \\ &= 0.879\,1. \end{aligned}$$

(b)

$$\begin{aligned} P(X = 5) &= \binom{10}{5} \left(\frac{1}{5}\right)^5 \left(\frac{4}{5}\right)^5 \\ &= 28 \left(\frac{1}{5}\right)^5 \left(\frac{4}{5}\right)^5 \\ &= 0.02642. \end{aligned}$$

(c)

$$\begin{aligned} P(X=10) &= \left(\frac{1}{5}\right)^{10} \\ &= 1.024 \times 10^{-7}. \end{aligned}$$

(d)

$$\begin{aligned} E(X) &= 10 \times \frac{1}{5} = 2, \\ \mathrm{Var}(X) &= \frac{8}{5}. \end{aligned}$$

15. If X is the number of meteorites hitting Mars per month, we are told $X \sim Po(6)$.

(a)

$$\begin{aligned} P(X \geq 2) &= 1 - P(X \leq 1) \\ &= 1 - \frac{6e^{-6}}{1} - e^{-6} \\ &= 0.982\,6. \end{aligned}$$

(b) If Y is the number of meteorites hitting Mars per six months, $Y \sim Po(36)$. So

$$P(Y=10) = \frac{e^{-36}36^{10}}{10!} = 2.337 \times 10^{-7}.$$

(c) If Z is the number of meteorites hitting Mars per year, $Z \sim Po(72)$, so $E(Z) = 72$.

16. .

(a) 0.1115

(b) 0.1645

(b) 0.0255

17. If X is the number of trials at which the first time the three coins show the same face, then $X \sim Ge(p)$ where

$$p = 2 \times \frac{1}{8} = \frac{1}{4}.$$

(a)

$$P(X > 3) = \left(1 - \frac{1}{4}\right)^3 = 0.4219.$$

(b) $E(X) = 4$.

BD